

Brief Report + AI Chat

Person 4: River Online Learning + ADWIN Drift Detection

Brief Explanation

Person 4 evaluates the retrieval system as a streaming task. The retrieval results are processed query by query, simulated feedback is created, and a River online model is updated after each prediction. Prequential evaluation is used, meaning the model predicts first, then learns from the actual feedback. ADWIN is used to monitor the prediction-error stream and detect possible drift.

Inputs

- **retrieval file** - retrieval results from Person 2, including ranked chunks and retrieval scores.
- **best_config.yaml** - best configuration from Person 3, used to initialize the hybrid weight/configuration.

Main Steps

- Load the retrieval file and the best configuration file.
- Convert ranked retrieval results into a query-level online stream.
- Generate simulated helpful/not helpful feedback from the retrieval outcome.
- Apply River online learning using prequential evaluation.
- Use ADWIN to detect drift in the prediction-error stream.
- Run an additional artificial drift experiment because the real stream is small and stable.

Outputs

- **online_learning_results.csv** - step-by-step predictions, feedback, accuracy, F1 score, alpha, and drift flags.
- **drift_events.csv** - ADWIN drift detections for the real retrieval stream, if any.
- **prequential_plot.png** - prequential performance plot over the query stream.
- **adaptive_alpha_plot.png** - plot of the adaptive hybrid weight over time.
- **artificial_drift_results.csv** and **artificial_drift_events.csv** - validation outputs for the controlled artificial drift experiment.
- **artificial_prequential_plot.png** - plot showing the injected drift point and ADWIN detection.
- **person4_summary.csv** / summary text - compact summary of the final results.

Important Note

The artificial drift experiment is clearly separated from the real experiment. It is included only to prove that ADWIN works correctly when a controlled drift is injected. The real retrieval stream may show no drift because it contains only 20 stable queries.

AI Chat Link

<https://chatgpt.com/share/6a09d2c0-6ed0-83eb-9fec-b9e9e35ae796>